Information Geometry of Biological Memory: How Slow Oscillations Carry High-Dimensional Programs

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Abstract

We present an information-geometric framework analyzing **phase-amplitude coupling** in nested oscillatory systems. The core contribution is an explicit observation model and Fisher-rank analysis: when slow phases modulate fast-band amplitude, they enter the fast-band likelihood (nonzero score). We prove that identifying $K-1$ relative phases from K coupled bands requires at least $K-1$ informative observations—coupling is necessary but not sufficient for identifiability. Global clock synchronization, by contrast, constrains the system to a one-dimensional submanifold, collapsing Fisher rank to at most one regardless of observation count.

We further discuss how **Takens delay embedding** can enable low-frequency carriers to encode high-dimensional trajectories—provided the slow waveform reflects genuine multidimensional dynamics rather than a pure sinusoid. This connection to delay embedding suggests (but does not prove) mechanisms by which biological delay architectures (cerebellar parallel fibers, thalamocortical loops) might decode slow-wave programs.

The mathematical results (Sections 2–6) are proved; the biological applications (Sections 7–8) are speculative hypotheses requiring empirical validation.

Keywords: Fisher information, Takens embedding, phase-amplitude coupling, biological memory, motor control, nested oscillations

1 Introduction

A central puzzle in biological computation: how do systems store and execute complex, high-dimensional programs using simple, low-frequency carriers?

Motor programs involve precise coordination across thousands of muscles on millisecond timescales. Memories encode rich, high-dimensional patterns of experience. Yet both rely heavily on slow oscillations—delta and theta rhythms that appear almost trivially simple when measured instantaneously. The cerebellum, hippocampus, and neocortex all exhibit

nested frequency hierarchies where slow rhythms modulate faster activity [2]. This architecture is not incidental; it appears ubiquitous across biological systems from neural circuits to circadian-ultradian metabolic cycles [10, 7].

We present a unified information-geometric framework explaining why this architecture works. Two mechanisms combine:

1. **Phase-amplitude coupling (PAC)** makes slow phases *identifiable* from fast observations. When gamma amplitude depends on theta phase, observing gamma bursts provides information about theta—the slow variable "enters the likelihood" of the fast signal.

2. **Takens embedding** allows low-frequency signals to *carry* high-dimensional structure—provided the slow waveform reflects genuine high-dimensional dynamics (not a pure sinusoid). Delay coordinates applied to such signals can reconstruct the underlying state space.

Two types of "dimension." This paper discusses two distinct notions of dimensionality that should not be conflated:

- **Phase-manifold dimension** ($\leq K-1$): The number of identifiable relative phase offsets between K frequency bands. This is the parameter space dimension in our Fisher-information analysis. For typical neural systems with 4–6 bands, this is 3–5 dimensions.

- **Program/state dimension** (potentially large): The dimension of the underlying dynamical system whose trajectory is encoded in slow-wave variations. This is what Takens embedding recovers, and can be much larger than $K-1$.

The PAC identifiability results (Sections 3–6) concern phase-manifold dimension. The Takens discussion (Section 7.4) concerns program/state dimension. These are complementary: PAC makes the phase coordinates identifiable; Takens explains how those coordinates can index high-dimensional programs.

Recent theoretical work proposes that intelligence requires maintaining high-dimensional coherent dynamics [12]. This paper provides the geometric foundation: nested frequencies are how biological systems create, store, and retrieve the high-dimensional structure that cognition requires.

1.1 Main Contributions

We develop an information-geometric framework with an explicit observation model, proving three results:

1. **Phase space structure** (Lemma 2): K oscillatory bands with unobservable global phase yield a $(K-1)$-dimensional parameter space $\mathbb{T}^{K-1}$ of relative phases.

2. **Coupling activates likelihood dependence** (Theorem 13): Under phase-amplitude coupling, slow phases enter the fast-band likelihood (nonzero score). Each observation carries joint information about multiple phase parameters, creating statistical dependence across directions. With sufficiently rich observations, this can raise Fisher rank toward $K-1$ (Proposition 8).

3. **Clock collapse** (Theorem 17): Global clock synchronization constrains relative phases to a 1-dimensional submanifold, collapsing Fisher rank to at most 1.

The key insight is that *nesting*—where slow phases modulate fast dynamics—creates statistical dependencies that make slow phases *present in* the fast-band likelihood. This is necessary but not sufficient for identifiability: one also needs observations rich enough to span the parameter directions. We formalize both conditions.

1.2 Notation

We use the following notation throughout:

- $\phi_k \in S^1$: phase of the k-th oscillatory band, with $\omega_1 < \omega_2 < \cdots < \omega_K$ (slow to fast)
- $\boldsymbol{\phi} = (\phi_1, \ldots, \phi_K) \in \mathbb{T}^K$: full phase vector
- $\psi_j = \phi_{j,0}$ for $j = 1, \ldots, K-1$: initial phase offsets of slow bands (parameters)
- $\boldsymbol{\psi} = (\psi_1, \ldots, \psi_{K-1}) \in \mathbb{T}^{K-1}$: parameter vector
- $\boldsymbol{y} = (y_1, \ldots, y_N)$: vector of N observations
- $p(\boldsymbol{y} \mid \boldsymbol{\psi})$: likelihood of observations given parameters
- $\mathcal{I}(\boldsymbol{\psi})$: Fisher information matrix at $\boldsymbol{\psi}$

Phase dynamics. Each oscillator has time-varying phase $\phi_k(t) = \omega_k t + \phi_{k,0}$, where ω_k is the (known) frequency and $\phi_{k,0}$ is the (unknown) initial phase offset. The *parameters* we wish to estimate are the initial phase offsets of slow bands:

$$\psi_j := \phi_{j,0}, \quad j = 1, \ldots, K-1 \tag{1}$$

under the gauge convention below. The instantaneous slow phase at time t is $\phi_j(t) = \omega_j t + \psi_j$.
Gauge convention. We fix the carrier phase $\phi_{K,0} = 0$. Under this gauge, only the slow-band offsets $\psi_1, \ldots, \psi_{K-1}$ are free parameters. This convention aligns with standard neuroscience practice: the fast carrier phase (e.g., gamma) is extracted via Hilbert transform and serves as the timing reference; slow phases (e.g., theta) are inferred from the amplitude modulation pattern.

2 Preliminaries

2.1 Statistical Manifolds and Fisher Information

A *statistical manifold* is a smooth family of probability distributions $\{p(y \mid \theta) : \theta \in \Theta\}$ parameterized by $\theta \in \Theta \subseteq \mathbb{R}^n$. The Fisher information matrix at θ is:

$$\mathcal{I}_{ij}(\theta) = \mathbb{E}_{y \sim p(\cdot|\theta)} \left[\frac{\partial \log p(y|\theta)}{\partial \theta^i} \frac{\partial \log p(y|\theta)}{\partial \theta^j} \right] \tag{2}$$

where the expectation is over observations y drawn from $p(y|\theta)$.

The *Fisher rank* at θ is $\operatorname{rank}(\mathcal{I}(\theta))$: the number of locally identifiable parameter directions. If $\mathcal{I}$ has a zero eigenvalue in direction v, then parameters cannot be distinguished along v from observations—the model is *unidentifiable* in that direction.

Remark 1. *Throughout this paper, Fisher information is always computed with respect to* parameters θ *(here, relative phases* $\boldsymbol{\psi}$*) in a likelihood* $p(y|\theta)$*, not with respect to coordinates of a distribution on phase space. This distinction is essential.*

2.2 Phase Variables and the Torus

A system with K oscillators has phase variables $\boldsymbol{\phi} = (\phi_1, \ldots, \phi_K) \in \mathbb{T}^K$, where $\mathbb{T}^K = (S^1)^K$ is the K-torus. In many settings, only *relative* phases are of physical interest—the absolute phase of any single oscillator has no intrinsic meaning without an external reference.

Lemma 2 (Gauge-fixed parameterization)**.** *The* K*-dimensional phase space* $\mathbb{T}^K$ *can be parameterized by:*

(i) A reference phase $\phi_{K,0} \in S^1$ *(the carrier initial offset, which we set to 0 by gauge choice), and*

(ii) $K-1$ *slow-band initial offsets* $\psi_j = \phi_{j,0}$ *for* $j = 1, \ldots, K-1$*.*

Under this gauge, $\boldsymbol{\psi} \in \mathbb{T}^{K-1}$ *are the free parameters.*

Proof. The initial phase vector $(\phi_{1,0}, \ldots, \phi_{K,0}) \in \mathbb{T}^K$ has K components. Fixing $\phi_{K,0} = 0$ leaves $K-1$ free parameters $(\psi_1, \ldots, \psi_{K-1}) \in \mathbb{T}^{K-1}$. □

Remark 3 (Why gauge-fixing works)**.** *Our observation model conditions on the carrier phase* $\phi_K(t) = \omega_K t$ *being extractable (e.g., via Hilbert transform of the bandpassed signal). Once the carrier phase is known, only the slow-band offsets* ψ_j *affect the amplitude modulation pattern. Shifting all phases by a global constant would change* $\phi_K(t)$*, but since we extract* $\phi_K(t)$ *from data rather than assuming it known* a priori*, this is operationally a gauge choice: we define "time zero" as a gamma peak.*

2.3 Observation Model: Phase-Amplitude Coupling

We now specify the generative model that makes "nested frequency hierarchy" statistically meaningful.

Definition 4 (Nested frequency observation model)**.** *A* nested frequency observation model *consists of:*

1. *Latent phases* $\phi_k(t) = \omega_k t + \phi_{k,0}$ *with frequencies* $\omega_1 < \omega_2 < \cdots < \omega_K$ *(slow to fast). Band* K *is the fast* carrier*; bands* $1, \ldots, K-1$ *are slow* modulators*.*

2. *A signal* $y(t)$ *of the form:*

$$y(t) = A(\phi_1(t), \ldots, \phi_{K-1}(t)) \cdot g(\phi_K(t)) + \epsilon(t) \tag{3}$$

   where $A(\cdot)$ *is the amplitude modulation depending on slow phases,* $g(\cdot)$ *is the fast carrier (e.g.,* $g(\phi) = \cos(\phi)$*), and* $\epsilon(t) \sim \mathcal{N}(0, \sigma^2)$ *is observation noise.*

3. *Under the gauge* $\phi_{K,0} = 0$*, the parameters are* $\boldsymbol{\psi} = (\psi_1, \ldots, \psi_{K-1})$ *where* $\psi_j = \phi_{j,0}$*.*

The critical feature is that **slow phases modulate the amplitude of fast oscillations**. The amplitude envelope $A(t)$ varies at the *slow* frequencies $\omega_1, \ldots, \omega_{K-1}$, while the carrier $g(\phi_K(t))$ oscillates at the fast frequency ω_K. This is the defining characteristic of phase-amplitude coupling.

Example 5 (Theta-gamma coupling)**.** *In hippocampal recordings, gamma oscillations (∼40 Hz) have amplitude modulated by theta phase (∼8 Hz) [9]. With* $K = 2$ *bands: band 1 = theta (slow modulator,* $\omega_\theta \approx 8$ *Hz), band 2 = gamma (fast carrier,* $\omega_\gamma \approx 40$ *Hz). The signal model is:*

$$y(t) = [1 + \alpha \cos(\omega_\theta t + \psi)] \cos(\omega_\gamma t) + \epsilon(t) \tag{4}$$

where $\psi = \phi_{\theta,0}$ *is the theta phase offset (the single parameter under gauge* $\phi_{\gamma,0} = 0$*), and* $\alpha \in [0, 1]$ *is the modulation depth. The amplitude envelope* $[1 + \alpha \cos(\omega_\theta t + \psi)]$ *varies at the* slow *theta frequency (∼8 Hz), while the carrier* $\cos(\omega_\gamma t)$ *oscillates at the* fast *gamma frequency (∼40 Hz). When* $\alpha = 0$*, theta phase does not affect the observation; when* $\alpha > 0$*, the modulation pattern depends on* ψ*.*

Remark 6 (Raw signal vs. extracted envelope)**.** *Our observation model uses the* raw *oscillatory signal* $y(t)$*. In practice, PAC analyses often first extract the amplitude envelope* $\hat{A}(t)$ *via Hilbert transform of the bandpassed fast signal, then study how* $\hat{A}(t)$ *covaries with slow phase.*

The identifiability arguments apply to both approaches:

- ***Raw signal****: Inference uses* $y(t_n)$ *directly; the carrier phase appears in the observation.*

- ***Extracted envelope****: If demodulation is perfect, the model becomes* $\hat{A}(t) = A(t; \boldsymbol{\psi}) + \eta(t)$*, where* η *is estimation noise. The slow phases still enter the likelihood through* $A(t; \boldsymbol{\psi})$*.*

The key point—that PAC makes slow phases enter the likelihood—holds in either case. We use the raw model for explicitness; envelope-based analyses are an implementation choice.

2.4 Multi-Sample Observations and Fisher Rank

A critical subtlety: for a *single* scalar observation y with Gaussian noise, the Fisher information matrix for any mean-parameterized model has the form

$$\mathcal{I}(\boldsymbol{\psi}) = \frac{1}{\sigma^2} \nabla_{\boldsymbol{\psi}} \mu \, (\nabla_{\boldsymbol{\psi}} \mu)^\top \tag{5}$$

which is an **outer product with rank at most 1**, regardless of parameter dimension. This means a single observation can identify at most one parameter direction.

To achieve higher Fisher rank, we need multiple observations. Let $\boldsymbol{y} = (y_1, \ldots, y_N)$ be N **independent** observations at times $t_1, \ldots, t_N$, with means $\mu_n(\boldsymbol{\psi}) = \mu(t_n; \boldsymbol{\psi})$. Under the independence assumption (i.i.d. noise across samples), the Fisher information is:

$$\mathcal{I}(\boldsymbol{\psi}) = \frac{1}{\sigma^2} \sum_{n=1}^{N} \nabla_{\boldsymbol{\psi}} \mu_n \, (\nabla_{\boldsymbol{\psi}} \mu_n)^\top \tag{6}$$

This sum of outer products can achieve rank up to $\min(N, K-1)$, provided the gradient vectors $\{\nabla_{\boldsymbol{\psi}} \mu_n\}$ span a $(K-1)$-dimensional subspace.

Remark 7 (Extension to correlated noise)**.** *Real neural time series have temporally correlated noise. With a full covariance matrix Σ, the Fisher information becomes $\mathcal{I} = J^\top \Sigma^{-1} J$ where J is the Jacobian of means. If Σ is full-rank, identifiability still depends on the rank of J—the span of gradients. Strong temporal correlation or low-rank noise structure can reduce effective information, but the core rank arguments remain valid.*

Proposition 8 (Fisher rank from multi-sample observations)**.** *Consider N independent Gaussian observations $y_n \sim \mathcal{N}(\mu_n(\boldsymbol{\psi}), \sigma^2)$ where $\boldsymbol{\psi} \in \mathbb{R}^{K-1}$.*

(i) The Fisher information matrix (6) *has* $\operatorname{rank}(\mathcal{I}) \leq \min(N, K-1)$.

(ii) To identify all $K-1$ phase offsets, we need at least $N \geq K-1$ observations whose gradient vectors $\nabla_{\boldsymbol{\psi}} \mu_n$ span $\mathbb{R}^{K-1}$.

(iii) For the nested frequency model (3) *with $A(t; \boldsymbol{\psi}) = A_0 + \sum_{j=1}^{K-1} \alpha_j \cos(\omega_j t + \psi_j)$ and $g(\phi_K) = \cos(\omega_K t)$, the Fisher rank equals $K-1$ for generic sampling, provided:*

- *all modulation depths $\alpha_j \neq 0$ for $j = 1, \ldots, K-1$,*
- *$N \geq K-1$ observations at distinct times, and*
- *the sampling times $\{t_n\}$ avoid the measure-zero set where gradients fail to span.*

Proof. (i) Each term $\nabla \mu_n (\nabla \mu_n)^\top$ has rank 1; the sum of N rank-1 matrices has rank at most N. The rank is also bounded by the parameter dimension $K-1$.

(ii) Full rank requires $\operatorname{span}\{\nabla \mu_1, \ldots, \nabla \mu_N\} = \mathbb{R}^{K-1}$, which needs at least $K-1$ vectors.

(iii) The mean at time t_n is $\mu_n = A(t_n) \cos(\omega_K t_n)$, so:

$$\frac{\partial \mu_n}{\partial \psi_j} = -\alpha_j \sin(\omega_j t_n + \psi_j) \cos(\omega_K t_n)$$

Define the $N \times (K-1)$ gradient matrix G with entries $G_{nj} = \sin(\omega_j t_n + \psi_j)\cos(\omega_K t_n)$. The Fisher matrix has full rank iff $\mathrm{rank}(G) = K-1$.

Why the bad set has measure zero: Consider any $(K-1)\times(K-1)$ submatrix of G (selecting $K-1$ sampling times). Its determinant is a trigonometric polynomial in $(t_1, \ldots, t_{K-1})$, hence analytic. This determinant is *not* identically zero: choosing sampling times t_n that avoid carrier zeros ($\cos(\omega_K t_n) \neq 0$) and are sufficiently spread (so the $\sin(\omega_j t_n + \psi_j)$ terms don't align across all j), one obtains a nonsingular submatrix. Since a non-identically-zero analytic function has measure-zero zero set in $\mathbb{R}^{K-1}$, the "bad" sampling schedules where all $(K-1)\times(K-1)$ minors vanish form a measure-zero set. □

Remark 9 (Connection to amplitude modulation)**.** *The nested frequency observation model is analogous to amplitude modulation in communications: slow phases modulate the envelope of the fast carrier, and the slow state can be recovered by demodulating the carrier. Fisher information formalizes the capacity to perform this inference under noise.*

Remark 10 (Local vs. global identifiability)**.** *Full Fisher rank implies* local *identifiability: at a given $\boldsymbol{\psi}$, small perturbations are distinguishable. However, on the torus $\mathbb{T}^{K-1}$, global ambiguities remain:*

- ***Periodicity****: ψ_j and $\psi_j + 2\pi$ are equivalent.*
- ***Carrier symmetries****: If the carrier phase is not perfectly fixed, additional equivalences may arise (e.g., $\psi \leftrightarrow \psi + \pi$ under certain symmetries).*

Thus full Fisher rank guarantees local identifiability (up to periodicity), but the posterior may be multi-modal. For estimation, this typically requires either prior information or constraints to resolve global ambiguity.

3 Fisher Information Under Independence

We first establish a baseline: what is the Fisher information structure when oscillatory bands are statistically independent in the observation model?

Proposition 11 (Block-diagonal Fisher under independence)**.** *Suppose the likelihood factorizes as $p(y|\boldsymbol{\psi}) = \prod_{j=1}^{K-1} p_j(y_j|\psi_j)$ where y_j are independent observations of each relative phase. Then the Fisher information matrix is block-diagonal:*

$$\mathcal{I}(\boldsymbol{\psi}) = \mathrm{diag}(\mathcal{I}_1(\psi_1), \ldots, \mathcal{I}_{K-1}(\psi_{K-1})) \tag{7}$$

where $\mathcal{I}_j(\psi_j) = \mathbb{E}[(\partial_{\psi_j} \log p_j)^2]$.

Proof. Under independence, $\log p(y|\boldsymbol{\psi}) = \sum_j \log p_j(y_j|\psi_j)$. The score for ψ_j is $\partial_{\psi_j} \log p = \partial_{\psi_j} \log p_j$, which depends only on y_j. For $j \neq k$:

$$\mathcal{I}_{jk} = \mathbb{E}[\partial_{\psi_j} \log p \cdot \partial_{\psi_k} \log p] = \mathbb{E}[\partial_{\psi_j} \log p_j] \cdot \mathbb{E}[\partial_{\psi_k} \log p_k] = 0 \tag{8}$$

since $\mathbb{E}[\partial_\psi \log p] = 0$ for any likelihood. □

Remark 12 (Unidentifiable phases under partial observation)**.** *Under the independent model, if we only observe the fast band (e.g., gamma), then slow phases (e.g., theta) do not appear in the likelihood at all. The corresponding rows and columns of $\mathcal{I}$ are zero—these parameters are* completely unidentifiable*.*

This sets up the key question: how does phase-amplitude coupling change this picture?

4 Coupling Activates Identifiability

We now prove that phase-amplitude coupling generically creates likelihood dependencies across phase parameters and, crucially, can make previously unidentifiable parameters identifiable.

4.1 General Result

Theorem 13 (Coupling activates likelihood dependence)**.** *Consider the observation model* (3) *with Gaussian noise $\epsilon \sim \mathcal{N}(0, \sigma^2)$. Parameterize by initial phase offsets $\boldsymbol{\psi} = (\psi_1, \ldots, \psi_{K-1})$ as in* (1)*. Let the amplitude modulation have the form $A(t; \boldsymbol{\psi}) = A_0 + \sum_{j=1}^{K-1} \alpha_j f_j(\omega_j t + \psi_j)$ where f_j are non-constant smooth functions (e.g.,* cos*) and α_j are modulation depths.*

(i) If $\alpha_j = 0$ for all j, then slow phases do not appear in the likelihood: the score $\partial_{\psi_j} \log p(y|\boldsymbol{\psi}) = 0$ identically, and these directions lie in the null space of $\mathcal{I}$.

(ii) If $\alpha_j \neq 0$ for some j, then ψ_j enters the likelihood*: the score depends nontrivially on ψ_j. For a single scalar observation, $\mathcal{I}$ has the outer-product form* (5) *with rank 1. With sufficiently rich observations (Proposition 8), the direction ψ_j can become identifiable.*

(iii) Under generic coupling (multiple $\alpha_j \neq 0$), each observation mean $\mu(t; \boldsymbol{\psi})$ depends on multiple phase parameters simultaneously. In the given parameterization, this means the score components $\partial_{\psi_j} \log p$ and $\partial_{\psi_k} \log p$ are generically coupled (both depend on the same observation), and the Fisher matrix is not block-diagonal. This reflects the structural fact that each observation carries joint information about multiple phases*—a property that persists under reparameterization.*

Proof. For Gaussian noise, the log-likelihood for a single observation at time t is:

$$\log p(y|\boldsymbol{\psi}) = -\frac{1}{2\sigma^2}[y - A(t; \boldsymbol{\psi}) \cos(\omega_K t)]^2 + \text{const} \tag{9}$$

where $A(t; \boldsymbol{\psi}) = A_0 + \sum_j \alpha_j f_j(\omega_j t + \psi_j)$ depends on the slow phase offsets.

The score for ψ_j is:

$$\frac{\partial \log p}{\partial \psi_j} = \frac{1}{\sigma^2}[y - A\cos(\omega_K t)] \cdot \alpha_j f_j'(\omega_j t + \psi_j) \cdot \cos(\omega_K t) \tag{10}$$

(i) If $\alpha_j = 0$, then $\partial A / \partial \psi_j = 0$, so the score is identically zero.

(ii) If $\alpha_j \neq 0$, then $\partial A/\partial\psi_j = \alpha_j f_j'(\omega_j t + \psi_j) \neq 0$ generically. The single-observation Fisher information is:

$$\mathcal{I}(\boldsymbol{\psi}) = \frac{1}{\sigma^2}\nabla_{\boldsymbol{\psi}}\mu\,(\nabla_{\boldsymbol{\psi}}\mu)^\top \tag{11}$$

which has rank 1 (outer product). By Proposition 8, with $N \geq K-1$ generic observations, the sum (6) can achieve full rank.

(iii) The gradient $\nabla_{\boldsymbol{\psi}}\mu$ has multiple nonzero components when multiple $\alpha_j \neq 0$. The outer product $vv^\top$ for $v = (v_1, \ldots, v_m)$ has $(vv^\top)_{jk} = v_j v_k \neq 0$ when both $v_j, v_k \neq 0$. □

Interpretation. The theorem formalizes a key distinction: *coupling makes slow phases present in the fast-band likelihood* (necessary condition), but *identifiability requires sufficient observations* to span the parameter directions (sufficiency). A single observation can identify at most one direction; recovering all $K-1$ relative phases requires at least $K-1$ informative samples.

4.2 Worked Example: Theta-Gamma System

We now compute the Fisher information explicitly for the canonical two-band case ($K = 2$), illustrating how coupling activates likelihood dependence.

Example 14 (Fisher information for theta-gamma coupling)**.** *Consider the model* (4) *at a single time t:*

$$y = [1 + \alpha\cos(\omega_\theta t + \psi)]\cos(\omega_\gamma t) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2) \tag{12}$$

Here $\psi = \phi_{\theta,0}$ is the theta phase offset—the single parameter in this $K = 2$ system (under gauge $\phi_{\gamma,0} = 0$).

What is known vs. unknown. *The frequencies $\omega_\theta, \omega_\gamma$ and modulation depth α are known (estimated from the power spectrum). The observation time t is known. The carrier phase $\omega_\gamma t$ is known because we fix the gauge. The unknown is the* theta phase offset *ψ, which determines the timing of amplitude peaks relative to the carrier.*

Define $\Theta(t) := \omega_\theta t + \psi$ (the instantaneous theta phase). The mean is $\mu = (1+\alpha\cos\Theta(t))\cos(\omega_\gamma t)$.

The score is:

$$\frac{\partial \log p}{\partial \psi} = -\frac{1}{\sigma^2}[y - \mu] \cdot \alpha\sin(\Theta(t))\cos(\omega_\gamma t) \tag{13}$$

(the minus sign arises because $\partial\mu/\partial\psi = -\alpha\sin(\Theta)\cos(\omega_\gamma t)$).

The Fisher information (for this single observation) is:

$$\mathcal{I}(\psi; t) = \frac{\alpha^2\sin^2(\omega_\theta t + \psi)\cos^2(\omega_\gamma t)}{\sigma^2} \tag{14}$$

Key observations:

1. *When $\alpha = 0$ (no coupling): $\mathcal{I} = 0$. The parameter ψ* does not appear *in the likelihood—it is absent, not merely hard to estimate.*

2. *When $\alpha > 0$ (coupling present): $\mathcal{I} > 0$ for generic ψ, t. The theta phase* enters *the likelihood and can contribute to identifiability.*

3. *This is a $K = 2$ system with one parameter, so a single informative observation suffices for identifiability (Proposition 8 with $K - 1 = 1$).*

4. *Fisher information scales as α^2: stronger coupling yields more precise estimation (lower Cramér-Rao bound).*

5. *Observations at $\omega_\gamma t = \pi/2$ (carrier zero-crossings) or $\omega_\theta t + \psi = 0, \pi$ (modulation extrema) are uninformative.*

This example illustrates the core mechanism: phase-amplitude coupling makes the slow phase *present in* the fast-band likelihood. For $K = 2$, one parameter and one observation suffice; for $K > 2$, we need at least $K - 1$ informative observations to span the parameter space.

Example 15 (Explicit Fisher metric for $K = 3$)**.** *Consider $K = 3$ bands with slow modulators at frequencies ω_1, ω_2 and carrier at ω_3. The parameter space is $\mathbb{T}^2 = S^1 \times S^1$ with coordinates $\boldsymbol{\psi} = (\psi_1, \psi_2)$.*

With $N = 2$ observations at times t_1, t_2, the gradient matrix is $G \in \mathbb{R}^{2\times 2}$ with entries

$$G_{nj} = -\alpha_j \sin(\omega_j t_n + \psi_j) \cos(\omega_3 t_n).$$

The Fisher metric (as a Riemannian metric on $\mathbb{T}^2$) is

$$g_{jk}(\boldsymbol{\psi}) = \mathcal{I}_{jk}(\boldsymbol{\psi}) = \frac{1}{\sigma^2} \sum_{n=1}^{2} \frac{\partial \mu_n}{\partial \psi_j} \frac{\partial \mu_n}{\partial \psi_k} = \frac{1}{\sigma^2} (G^\top G)_{jk}. \tag{15}$$

Geometric interpretation. *The metric g defines distances on $\mathbb{T}^2$: the length of a curve $\boldsymbol{\psi}(s)$ is $\int \sqrt{g_{jk}\dot{\psi}^j \dot{\psi}^k}\, ds$. Directions with large eigenvalues are "stiff" (small parameter changes produce large likelihood changes); directions with small eigenvalues are "sloppy" (hard to identify).*

At generic (ψ_1, ψ_2) with well-chosen (t_1, t_2), the metric is positive-definite with rank 2, confirming both phases are locally identifiable. The metric varies smoothly over $\mathbb{T}^2$, making the statistical manifold a genuine Riemannian manifold.

4.3 What Coupling Does *Not* Do

To prevent misunderstanding, we clarify what Theorem 13 does *not* claim:

1. **Coupling does not increase dimension beyond the parameter space.** If you have $K - 1$ relative phases, the Fisher matrix is at most $(K - 1) \times (K - 1)$ with rank at most $K - 1$. Coupling cannot create dimensions that don't exist.

2. **A single scalar observation has Fisher rank at most 1.** Even with coupling, a single observation cannot identify more than one parameter direction (equation (5)). High-dimensional identifiability requires multiple observations whose gradients span the parameter space.

3. **Strong coupling can reduce effective dimension.** Phase-locking (complete synchronization) imposes constraints that *reduce* the number of free parameters. Our "activation" result concerns the regime where coupling creates informative dependencies, not hard constraints.

4. **Identifiability requires appropriate observations.** If your measurement is insensitive to amplitude modulation, coupling won't help. The result is model-dependent.

5 Clock Synchronization Collapses Dimension

We now characterize what happens when all oscillators are locked to a global clock.

Definition 16 (Clock synchronization)**.** *A system of K oscillators is* clock-synchronized *if all phases are integer multiples of a single base clock:*

$$\phi_k(t) = n_k \Phi(t), \quad \text{where } \Phi(t) = \omega_0 t + \Phi_0 \tag{16}$$

Here $\Phi(t)$ is the base clock phase, Φ_0 is the initial clock offset (unknown to the observer), ω_0 is the base frequency, and n_k are positive integer multipliers with $n_1 < n_2 < \cdots < n_K$.

Under clock synchronization, the instantaneous relative phases (with respect to the carrier, band K) become:

$$\delta_j(t) := \phi_j(t) - \phi_K(t) = (n_j - n_K)\Phi(t) \tag{17}$$

(We use δ_j rather than ψ_j to distinguish these time-varying locked phases from the free parameters ψ_j in the nested model.) All relative phases are deterministic functions of the single clock phase $\Phi(t)$.

Theorem 17 (Clock collapse)**.** *Under clock synchronization (Definition 16):*

(i) The relative phases $\delta_j(t)$ are deterministic functions of $\Phi(t)$. If observation time t is known, then $\Phi(t) = \omega_0 t + \Phi_0$ depends only on the single unknown Φ_0.

(ii) The effective parameter dimension is at most 1: the only free parameter is the clock offset Φ_0. The Fisher information for multi-sample observations has rank at most 1.

(iii) The trajectory $\boldsymbol{\delta}(\Phi)$ traces a 1-dimensional closed curve in $\mathbb{T}^{K-1}$. Since the coefficients $(n_j - n_K)$ are integers, the trajectory is periodic in Φ with period $2\pi/\gcd(|n_1 - n_K|, \ldots, |n_{K-1} - n_K|)$. The system is constrained to this curve; it cannot access the full $(K-1)$-dimensional torus.

Proof. (i) By definition, $\delta_j(t) = (n_j - n_K)\Phi(t)$. With t known, all δ_j are functions of the single parameter Φ_0.

(ii) The mean $\mu(t; \Phi_0)$ depends on Φ_0 through $\Phi(t)$. The Fisher information from N observations is:

$$\mathcal{I}(\Phi_0) = \frac{1}{\sigma^2} \sum_{n=1}^{N} \left(\frac{\partial \mu_n}{\partial \Phi_0} \right)^2 \tag{18}$$

This is a sum of scalars (each observation contributes to estimating the single parameter Φ_0), not a matrix—the parameter space is 1-dimensional.

(iii) The map $\Phi \mapsto ((n_1 - n_K)\Phi, \ldots, (n_{K-1} - n_K)\Phi)$ has 1-dimensional image in $\mathbb{T}^{K-1}$. Since all coefficients are integers, the image is a closed curve (torus knot), not a dense winding. □

Interpretation. A global clock constrains the system to a 1-dimensional submanifold of phase space. Instead of $K - 1$ independent relative phases, the clock imposes $(K - 2)$ constraints, leaving only the clock offset as a free parameter. No amount of observation can identify more than this single dimension—the constraint is geometric, not statistical.

Figure 1 illustrates this contrast for $K = 3$: the full parameter space is the 2-torus $\mathbb{T}^2$, but clock synchronization constrains the system to a 1-dimensional closed curve (a torus knot) winding through this space.

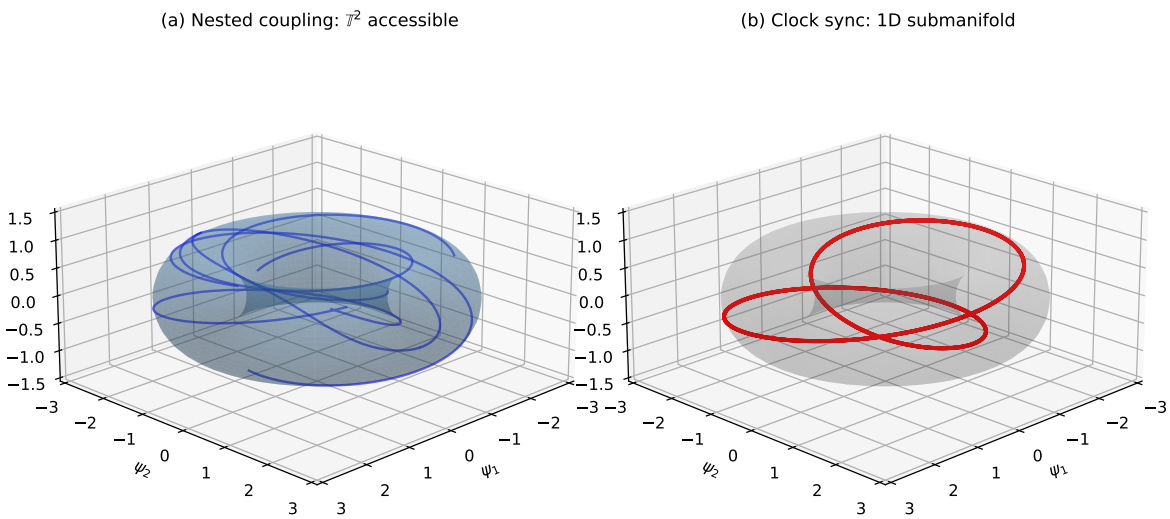

Figure 1: Geometric contrast between nested coupling and clock synchronization for $K = 3$ bands. (a) Under nested coupling, trajectories can explore the full 2-torus $\mathbb{T}^2$ of relative phase offsets (ψ_1, ψ_2). (b) Under clock synchronization, all relative phases are deterministic functions of a single base clock, constraining the system to a 1-dimensional closed curve (red). The clock curve cannot access most of the torus—this is the geometric origin of dimensional collapse.

Remark 18 (Comparing assumptions across models)**.** *To ensure a fair comparison between nested and clocked systems:*

- ***Both models****: observation times t_n are known; noise is i.i.d. Gaussian; frequencies are known.*
- ***Nested model****: carrier phase is extractable (gauge-fixed); parameters are slow-band offsets $\psi_j = \phi_{j,0}$ for $j = 1, \ldots, K - 1$, each freely varying.*

- ***Clock model***: *integer multipliers n_k are known; the single parameter is the clock offset Φ_0, which determines all phases simultaneously.*

The dimensional contrast arises from the constraint structure*, not from asymmetric assumptions about what is observed. Clock synchronization* deterministically couples *all phases (all are integer multiples of one base), collapsing $K-1$ potential degrees of freedom to 1. Nested coupling leaves slow-band offsets free (amplitude-modulated, but not phase-locked).*

5.1 Application to Digital Systems

Digital computers with a global clock operate in this regime: all state transitions are synchronized to clock edges, and relative timing between operations is fixed to integer clock cycles.

Remark 19 (Nuances for digital systems)**.** *The clock-collapse result applies to* ideal globally-clocked systems*. Real digital systems have complications:*

1. ***Multiple clock domains***: *Modern processors have multiple asynchronous clock domains, partially recovering independent phase relationships.*

2. ***Clock jitter***: *Thermal and quantum noise create phase uncertainty, though typically small.*

3. ***Asynchronous circuits***: *Clockless designs exist and do not suffer clock collapse.*

The contrast we emphasize is between systems designed around a global timing reference *versus* systems with intrinsically asynchronous, hierarchically coupled timing*. The former collapse relative phase degrees of freedom; the latter preserve them.*

Remark 20 (Dimension in digital systems)**.** *Digital systems have enormous* discrete *state spaces (exponential in bit count). Theorem 17 concerns* continuous relative phase *degrees of freedom, not discrete state dimension. A clocked digital computer gains combinatorial complexity in bits but loses the kind of continuous multiscale phase manifold that characterizes biological oscillatory systems.*

6 Summary of Statistical Structure

We summarize the Fisher information structure under different architectures:

Corollary 21 (Identifiability gap)**.** *Under the nested frequency observation model (Definition 4) with $N \geq K-1$ observations:*

1. *With K coupled bands and generic modulation depths, up to $K-1$ relative phases can become identifiable from fast-band observations alone (Proposition 8).*

2. *Under global clock synchronization, at most 1 parameter (clock offset) is identifiable, regardless of observation count.*

Architecture	Parameter space	Fisher structure	Identifiable dim.
K independent bands	$\mathbb{T}^{K-1}$	Factorized likelihood	$\leq K-1^{\dagger}$
K nested bands (coupled)	$\mathbb{T}^{K-1}$	Non-factorized likelihood	$\leq K-1^{*}$
Clock-synchronized	1D submanifold	Scalar	≤ 1

Table 1: Fisher information structure under different oscillatory architectures. †Requires direct observation of each band. *With $N \geq K-1$ informative observations; coupling makes slow phases enter the fast-band likelihood.

The maximal identifiability gap is $K-2$ dimensions: nested coupling can support $K-1$ identifiable directions where clock synchronization supports only 1.

For neural systems with 4–6 canonical frequency bands (delta, theta, alpha, beta, gamma, high-gamma), this represents up to 3–5 identifiable relative phases under coupling with sufficient observations, versus at most 1 under global clock synchronization.

Figure 2 illustrates this contrast with a simulation: for $K = 4$ bands (three slow modulators at 2, 5, 8 Hz; one carrier at 40 Hz), the nested model achieves full rank ($K-1=3$) with $N \geq 3$ observations, while the clock model remains at rank 1 regardless of N.

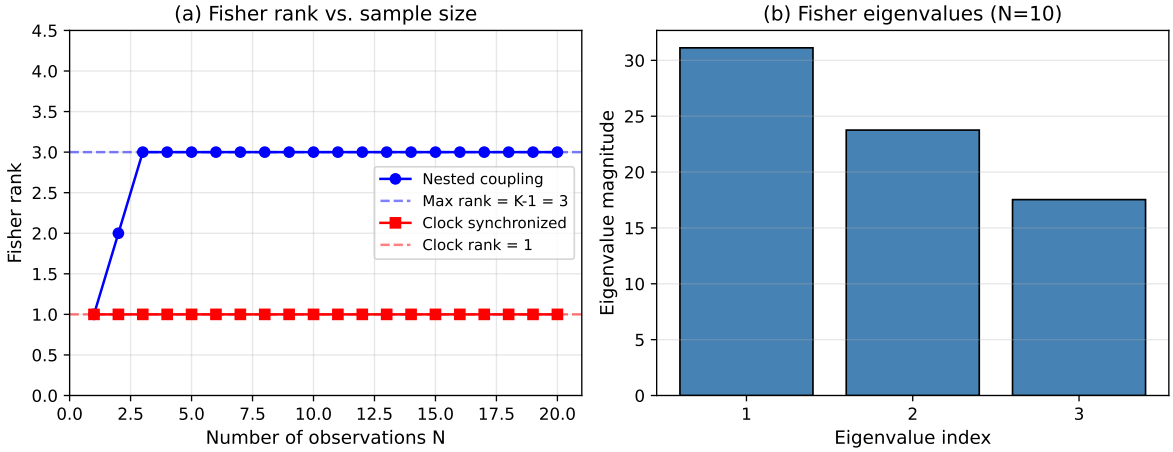

Figure 2: Fisher rank under nested coupling vs. clock synchronization. (a) Rank of Fisher information matrix as a function of sample size N. The nested model achieves full rank $K-1=3$ once $N \geq K-1$; the clock model is constrained to rank 1 regardless of N. (b) Eigenvalue spectrum for the nested model with $N = 10$ observations, showing three nonzero eigenvalues corresponding to the three identifiable slow-phase parameters. **Simulation parameters:** $K = 4$ bands with slow modulators at 2, 5, 8 Hz and carrier at 40 Hz; modulation depths $\alpha_j = 0.3$; noise $\sigma = 0.1$; observations uniformly sampled on $[0, 1]$ s with small random jitter.

7 Discussion

Scope note. Sections 2–6 contain the formal results: definitions, theorems, and proofs about Fisher information structure under nested coupling vs. clock synchronization. This section and Section 8 (Applications) offer *interpretation and speculation*—connections to Takens embedding, biological mechanisms, and clinical hypotheses that are suggested by the formalism but not proved by it. We flag speculative claims explicitly.

7.1 What Nesting Contributes Beyond Multiple Oscillators

A reviewer might ask: "Any K oscillators give a K-torus; what is special about nesting?"

The answer is in the observation model. *Nesting* refers to the hierarchical structure where slow phases modulate fast dynamics:

1. **Slow phases are not directly observed**—in typical recordings, you measure fast activity (spikes, gamma, etc.), not the slow envelope directly.

2. **Phase-amplitude coupling makes slow phases enter the fast-band likelihood**—this is the "activation" of Theorem 13. Without coupling, the slow-phase score is identically zero.

3. **Multi-sample observations can achieve full-rank Fisher information**—Proposition 8 shows that with $N \geq K - 1$ informative observations, coupling enables identification of all $K - 1$ relative phases from fast-band data alone.

4. **The hierarchy provides natural multiscale coordinates**—slow phases set context, fast phases carry content, and the nesting structure determines which variables can be inferred from which observations.

Without the nesting structure (phase-amplitude coupling), independent oscillators would give a product space with diagonal (or zero) Fisher information—you could estimate phases you directly observe, but not others. Nesting creates the statistical dependencies that make the full phase structure accessible from partial observations.

7.2 Why Biology Uses Nested Frequencies

The ubiquity of nested frequency hierarchies in biological systems [2, 9] is often attributed to functional requirements: temporal segregation, multiplexing, context-dependent processing.

Our analysis suggests a complementary geometric explanation: **nested frequencies are how biological systems create identifiable multiscale structure**. The slow contextual variables (mood, task state, circadian phase) become estimable from fast observables (neural spikes, gene expression bursts) precisely because of cross-scale coupling.

This may explain why biological systems consistently employ nested architectures rather than independent oscillators: the nesting structure maximizes what can be inferred about system state from available observations.

7.3 The Phase Manifold as Continuous Timing Structure

The Fisher metric on $\mathbb{T}^{K-1}$ provides a *continuous* coordinate system for relative timing relationships. Phase offsets vary continuously, and the statistical dependencies created by PAC are smooth functions of these offsets.

This contrasts with globally clocked systems, where timing advances in discrete integer steps. The continuous phase manifold may be relevant to biological coordination problems (motor control, perception, memory) that require smooth interpolation of timing relationships—though formalizing spatial aspects (conduction delays, propagation topology) requires extending the present framework.

7.4 Takens Embedding: Low Frequency Carries High Dimension

A natural question: if slow oscillations are "low-frequency," how can they carry complex, high-dimensional programs?

Takens' embedding theorem [13] provides the answer. For a deterministic dynamical system with d degrees of freedom, time-delayed copies of a single scalar observable—$x(t), x(t-\tau), x(t-2\tau), \ldots$—can reconstruct the system's complete state space topology. Specifically, an embedding dimension of $2d+1$ delay coordinates generically suffices for diffeomorphic reconstruction.

Formal assumptions. The theorem requires: (i) smooth deterministic dynamics on a compact manifold, (ii) a generic (non-degenerate) scalar observation function, and (iii) sufficient embedding dimension. In practice, neural data involves noise and nonstationarity, so embeddings are approximate. There is a substantial literature on delay embedding with noise [4]; the key insight is that moderate noise degrades but does not destroy the topological structure, while embedding dimension may need to be increased.

Critical caveat. If the slow signal were an exact sinusoid, delay embedding would recover only a **2-dimensional closed curve** (an ellipse)—not a high-dimensional attractor. High-dimensional reconstruction requires that the observed slow waveform is a *generic scalar measurement of a higher-dimensional state*. In biological oscillations, this condition is satisfied because:

- Amplitude varies with system state (arousal, attention, task demands)
- Frequency drifts and phase slips encode state transitions
- Non-sinusoidal waveform components (harmonics) carry additional information
- Nonstationarity reflects the trajectory through state space

The "slow oscillation" is not a pure sinusoid but a *projection* of high-dimensional neural dynamics onto a narrow frequency band. It is this projection, not the carrier frequency per se, that determines embedding dimension.

The implication: **low frequency does not mean low complexity**, provided the slow signal reflects genuine high-dimensional dynamics rather than an isolated oscillator. The complexity is encoded in the *waveform variations*, not the carrier frequency.

Biological delay architectures can implement Takens embedding in hardware. Consider a delta-band signal (∼2 Hz) with appropriate delay coordinates:

$$\mathbf{x}(t) = \big[y(t),\, y(t-\tau),\, y(t-2\tau),\, \ldots\big] \tag{19}$$

The optimal embedding delay is $\tau^* \approx 1/(4f)$ [14], placing delay taps at quarter-period intervals. For a 2 Hz delta oscillation, $\tau^* \approx 125$ ms; for 40 Hz gamma, $\tau^* \approx 6$ ms. Cerebellar parallel fibers (delays 2–8 ms) and thalamocortical loops (delays 10–100 ms) operate in precisely these ranges.

7.4.1 Formal Connection: Delay Vectors as Multi-Sample Observations

The connection between Takens embedding and Fisher rank is not merely analogical—it is exact. A delay vector *is* a multi-sample observation in the sense of Proposition 8.

Proposition 22 (Delay embedding and Fisher rank)**.** *Consider the nested frequency model* (3) *observed via an m-dimensional delay vector:*

$$\mathbf{y}(t) = \big(y(t), y(t-\tau), y(t-2\tau), \ldots, y(t-(m-1)\tau)\big)^\top \tag{20}$$

where each component $y(t-k\tau) = A(t-k\tau)\cos(\omega_K(t-k\tau)) + \epsilon_k$ with independent noise $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$.

Under the conditions of Proposition 8, the Fisher information matrix for $\boldsymbol{\psi}$ based on observing $\mathbf{y}(t)$ is:

$$\mathcal{I}(\boldsymbol{\psi}) = \frac{1}{\sigma^2} \sum_{k=0}^{m-1} \nabla_{\boldsymbol{\psi}}\mu(t-k\tau)\,\big(\nabla_{\boldsymbol{\psi}}\mu(t-k\tau)\big)^\top \tag{21}$$

where $\mu(s) = A(s;\boldsymbol{\psi})\cos(\omega_K s)$ is the mean at time s.

If $m \geq K-1$ and the delay τ is chosen such that the gradient vectors $\{\nabla_{\boldsymbol{\psi}}\mu(t-k\tau)\}_{k=0}^{m-1}$ span $\mathbb{R}^{K-1}$, then $\operatorname{rank}(\mathcal{I}) = K-1$.

Proof. The delay vector has independent Gaussian components with means $\mu_k = \mu(t-k\tau)$. By the additivity of Fisher information for independent observations, the total Fisher matrix is the sum (21). The rank argument follows from Proposition 8: with $m \geq K-1$ delay taps at times that avoid the measure-zero bad set, the gradients span the parameter space. □

Biological interpretation. Neural delay lines (axonal conduction, synaptic delays) implement (21) in hardware. A neuron receiving inputs via parallel fibers with staggered delays effectively computes a delay vector. The downstream readout performs implicit Fisher-optimal inference on $\boldsymbol{\psi}$—the slow-phase structure encoded in the signal.

Optimal delay. For oscillatory signals, the optimal embedding delay is $\tau^* \approx 1/(4f_{\min})$ where $f_{\min}$ is the lowest frequency of interest [14]. This places delay taps at quarter-period intervals, maximizing the angle between successive gradient vectors and hence the Fisher rank. Cerebellar parallel fibers (delays 2–8 ms) and thalamocortical loops (delays 10–100 ms) operate in ranges appropriate for beta/gamma and theta/delta bands respectively.

7.5 Control Leverage: Where Information Injection Matters

A critical corollary of Takens embedding: **where you inject information determines how much dimensionality you affect**.

The slow-frequency carrier IS the compressed high-dimensional program. Via Takens embedding, a perturbation to the slow waveform trajectory changes the *entire* reconstructed state-space path. Injecting information at the low-frequency level gives access to high-dimensional outcomes—you are modifying the program, not just one moment of execution.

By contrast, injecting information at high frequencies (gamma) affects only local, momentary output. Gamma is the readout layer—low-dimensional execution of whatever program the slow frequencies encode.

Injection point	**What is modified**	**Downstream effect**
Low-F (delta/theta)	Carrier trajectory	Entire high-D program changes
High-F (gamma)	Execution/readout	Local, momentary effect only

This has immediate implications:

- **Memory consolidation during sleep:** Writing to slow-wave carriers modifies the high-dimensional programs themselves, not just their readout.
- **Motor learning:** Cerebellar plasticity at the delay-line level changes which programs get reconstructed from which slow-wave trajectories.
- **Metabolic protection:** Slow-wave infrastructure is the high-leverage control layer; losing it means losing control over downstream dimensionality. This explains why biological systems invest heavily in maintaining slow-wave generation.
- **Therapeutic targeting:** Interventions at gamma frequencies cannot restore missing program structure; only supporting slow-wave infrastructure can restore high-dimensional coordination.

7.6 Implications for Artificial Systems

If high-dimensional identifiable structure requires nested frequency architectures with phase-amplitude coupling, then achieving similar capabilities in artificial systems may require:

1. Multiple oscillatory timescales (not a single global clock)
2. Asynchronous phase relationships between scales
3. Coupling mechanisms that make slow state visible in fast dynamics

Neuromorphic computing architectures that incorporate these features [8, 11] may achieve richer statistical structure than globally clocked systems. The key is not merely parallelism or speed, but the geometric structure of timing relationships.

8 Applications

The framework has implications across neuroscience, clinical medicine, and behavioral science.

8.1 Motor Control and the Cerebellum

The cerebellum's parallel fiber architecture may implement Takens embedding in hardware. Parallel fibers conduct at approximately 0.3–0.5 m/s [5], creating delays of several milliseconds across the granular layer—a range appropriate for embedding signals in the beta/gamma band relevant for motor timing.

The cerebellum as Takens decoder: Motor cortex generates slow-wave programs encoding complex motor sequences. The cerebellum receives these programs and, via delay-line architecture, reconstructs the full state-space trajectory. This explains why the cerebellum is essential for skilled movement: it unpacks the compressed program into real-time motor commands.

Chair flying / motor imagery: Mental rehearsal of motor programs works because the slow-wave program can run through the cerebellar decoder without executing the movement. The Takens reconstruction is performed; only the final output is inhibited. This provides a mechanism for "offline" motor learning.

8.2 Memory Consolidation and Sleep

Slow-wave sleep is dominated by delta oscillations that carry compressed memory programs:

Hippocampal replay: During sleep, the hippocampus replays compressed sequences at time scales much faster than original experience. This "replay" is Takens reconstruction—delay coordinates applied to slow oscillations recover the full episodic structure.

Why sleep consolidates memory: Memory transfer from hippocampus to neocortex requires the receiving circuits to sample the slow-wave trajectory at appropriate delays. This cannot be rushed; the information *is* the temporal unfolding. Sleep provides the time for complete Takens reconstruction.

Why dreams are fragmentary: Partial or noisy Takens reconstruction from incomplete slow-wave programs produces the characteristic fragmented, recombinant quality of dream experience.

8.3 Clinical Hypotheses: Dimensional Collapse in Pathology

The framework suggests testable hypotheses about pathological states characterized by reduced neural complexity:

Hypothesis (motor neuron disease): If motor coordination depends on high-dimensional PAC structure, then degeneration of cortical interneurons that maintain slow-wave infrastructure could reduce Fisher rank below the threshold needed for coordinated motor output. This predicts that PAC degradation should precede motor symptoms—a testable prediction.

Hypothesis (depression): If reduced neural complexity accompanies depression (as some EEG studies suggest, though evidence is mixed [6]), the framework offers one possible

mechanism: degraded slow-wave scaffolding → reduced PAC → loss of high-dimensional cognitive flexibility. If correct, interventions that restore oscillatory structure (exercise, rhythmic activity, sleep hygiene) should show corresponding PAC normalization. This remains speculative pending systematic PAC measurements in depression.

These are speculative applications that require empirical validation. The core mathematical contribution—the Fisher rank analysis—stands independently of whether these clinical hypotheses prove correct.

8.4 Limitations

This analysis makes several idealizations:

1. **Idealized oscillator models**: Real biological oscillations have amplitude dynamics, non-sinusoidal waveforms, and complex coupling functions. The sinusoidal model captures the essential mechanism but not the full complexity.

2. **Gaussian noise**: The explicit calculations assume Gaussian observation noise; non-Gaussian cases may differ quantitatively (though the rank arguments remain valid).

3. **Known frequencies and coupling strengths**: We treat frequencies ω_k and modulation depths α_j as known. If these are unknown nuisance parameters, the augmented Fisher matrix has additional rows/columns, potentially reducing effective rank for the phase parameters of interest.

4. **Carrier phase extraction**: The model conditions on the carrier phase being extractable via Hilbert transform. In practice, this introduces estimation error not captured here.

5. **Generic conditions**: Proposition 8(iii) assumes "generic" sampling. Pathological cases (e.g., sampling only at carrier zero-crossings) can reduce rank.

Despite these limitations, the core insight—that cross-scale coupling makes slow phases *present in* the fast-band likelihood, enabling identification with sufficient observations—should be robust across more realistic models.

9 Conclusion

We have presented an information-geometric framework analyzing how phase-amplitude coupling creates identifiable statistical structure in nested oscillatory systems.

The core contributions:

1. **Phase-amplitude coupling creates identifiability** (Theorem 13): Slow phases enter the fast-band likelihood, enabling statistical estimation. Without PAC, slow phases are absent from observations, not merely hard to estimate.

2. **Multi-sample observations achieve full rank** (Proposition 8): A single scalar observation has Fisher rank ≤ 1. With $N \geq K - 1$ generic observations, coupling enables identification of all $K - 1$ slow-band phases.

3. **Clock synchronization collapses dimension** (Theorem 17): Global clock locking constrains relative phases to a 1D submanifold, limiting Fisher rank to at most 1 regardless of observation count.

The identifiability gap: Biological systems with K nested coupled bands can achieve Fisher rank up to $K - 1$ (Corollary 21). Clock-synchronized systems are limited to rank 1. This gap of up to $K - 2$ dimensions is geometric—it reflects fundamental constraints on what can be identified from observations.

Connections to biological function: We suggest (but do not prove) that Takens delay embedding provides a mechanism by which neural delay architectures convert temporal structure into high-dimensional spatial coordinates, and that this may explain aspects of motor control, memory consolidation, and sleep. These applications are speculative and require empirical validation.

Whether artificial systems can achieve similar high-dimensional identifiable structure by adopting nested asynchronous architectures remains an open question for future work.

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